Prasanna Sattigeri

Principal Research Scientist and Manager, IBM Research · Principal Investigator, MIT-IBM Watson AI Lab

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SUMMARY

I lead post-training for agent reliability at IBM Research. The team trains Granite Guardian, an open judge model, and Granite Switch, a Granite model composed from adapter functions that check its own work. Research in post-training and generative modeling. 10,043 citations, h-index 44 (Google Scholar, October 2026).

EXPERIENCE

IBM Research and MIT-IBM Watson AI Lab · Principal Research Scientist and Manager

2015 – present

- Lead the Granite Guardian team through five releases in 18 months, from 3.0 (October 2024) to 4.1 (April 2026): training data, post-training, evaluation and open-source release.
- Granite Guardian 4.1 8B: fifth of 20 guardrail configurations on the Artificial Analysis benchmark (F1 0.859, 59 ms), first among models that take their policy at runtime. As a reward model, 70.29 on JETTS best-of-N selection, ahead of reward models up to 70B. Six of the top ten places on GuardBench (April 2025). Third of 39 on the LLM-AggreFact leaderboard.
- Lead post-training for Granite Switch: adapter functions for policy compliance, factuality detection and correction, requirement checking, calibrated uncertainty and context attribution. Requirement checking raises IFEval balanced accuracy from 0.569 (prompted base model) to 0.855.
- Adoption: used in IBM watsonx.ai and watsonx.governance, deployed as the guardrails in Red Hat's Ask Red Hat assistant, integrated in Traefik Hub, deepset Haystack and Mozilla.ai any-guardrail. 1.15M Hugging Face downloads and 371K Ollama pulls (October 2026).
- Principal investigator on two MIT-IBM Watson AI Lab projects, with Gregory Wornell and David Sontag. Thesis committee member for PhD students at MIT and RPI.
- Core contributor to AI Fairness 360, AI Explainability 360 and Uncertainty Quantification 360. Generative modeling research including DIP-VAE (ICLR 2018) and semi-supervised GANs (NeurIPS 2017).

Yelp Inc. · Software Engineer

2014 – 2015

- Machine learning for search, using natural language processing and image understanding.

Arizona State University · Research Assistant

2009 – 2014

- Sparse and low-rank representation learning; lead student researcher on an NSF sensor project.

EDUCATION

2014 Ph.D., Electrical Engineering, Arizona State University

2008 B.Tech., Electronics Engineering, National Institute of Technology, India

SELECTED PUBLICATIONS

Building a Foundational Guardrail for General Agentic Systems via Synthetic Data. *ICLR 2026, co-corresponding author*

Decocted Experience Improves Test-Time Inference in LLM Agents. *COLM 2026*

Answering the Wrong Question: Reasoning Trace Inversion for Abstention in LLMs. *ACL 2026, senior author*

Granite Guardian: Comprehensive LLM Safeguarding. *NAACL 2025, senior author*

Multi-Level Explanations for Generative Language Models. *ACL 2025, oral*

Thermometer: Towards Universal Calibration for Large Language Models. *ICML 2024*

Graph-based Uncertainty Metrics for Long-form Language Model Generations. *NeurIPS 2024*

Effective Human-AI Teams via Learned Natural Language Rules and Onboarding. *NeurIPS 2023, spotlight*

Fair Infinitesimal Jackknife: Mitigating the Influence of Biased Training Data Points Without Refitting. *NeurIPS 2022, first author*

Variational Inference of Disentangled Latent Concepts from Unlabeled Observations. *ICLR 2018*

Semi-supervised Learning with GANs: Manifold Invariance with Improved Inference. *NeurIPS 2017*

AWARDS

2025	IBM Research Outstanding Accomplishment: AI governance innovations to watsonx.governance and open source
2025	IBM Research Technical Achievement Award: Granite Guardian
2024	IBM Research Outstanding Accomplishment: Granite 3.0 release
2024	IBM Research Technical Achievement Award: Causal inference for business decision making
2020–21	Four IBM Research Technical Achievement Awards: Uncertainty quantification, robust and generalizable AI, learning with less labels, efficient AI
2019	IBM Outstanding Technical Achievement Award: Contributions to trustworthy AI

PATENTS AND SERVICE

- Seven granted US patents and six applications.
- Associate Editor, Pattern Recognition. Area chair and senior program committee: NeurIPS, ICML, ICLR, AAAI.
- IEEE Senior Member.